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Principal Component Analysis of Solar Flares in the Soft X-ray Flux

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Summary. Principal component analysis is a technique for extracting the salient features from a mass of data. It applies, in particular, to the analysis of nonstationary ensembles. Computational schemes for this task require the evaluation of eigenvalues of matrices. We have used EISPACK Matrix Eigen System Routines on an IBM 360-75 to analyze full-disk proportional-counter data from the X-ray event analyzer (X-REA) which was part of the Skylab ATM/S-056 experiment.

Empirical orthogonal functions have been derived for events in the soft X-ray spectrum between 2.5 and 20 Å during different time frames between June 1973 and January 1974. Results indicate that approximately 90% of the cumulative power of each analyzed flare is contained in the largest eigenvector. The first two largest eigenvectors are sufficient for an empirical curve-fit through the raw data and a characterization of solar flares in the soft X-ray flux. Power spectra of the two largest eigenvectors reveal a previously reported periodicity of approximately 5 min. Similar signatures were also obtained from flares that are synchronized on maximum pulse-height when subjected to a principal component analysis.

Key words: Sun – flares – X-rays – power spectra – principal components

I. Introduction

In a previous paper, based on the power-spectrum analysis of proportional counter data obtained using the X-ray event analyzer (X-REA), a part of the Skylab ATM/S-056 X-ray experiment (Underwood et al., 1977), we presented evidence for a 256-s oscillation in the solar, soft X-ray flux during the flare of 15 June 1973 (Teuber et al., 1978). No X-ray oscillations were apparent in the non-flaring solar data, leading us to suggest that X-ray oscillations may be associated only with flare-like activity. Wolff (1972) has suggested that oscillations of the Sun might be excited by a large solar flare, and Hénoux and Nakagawa (1978) investigated the dynamics of a model solar atmosphere irradiated by soft X-rays in a 1 N optical-class flare and found that the atmosphere undergoes an oscillating expansion characterized by a 270-s oscillation period. Other power-spectrum analysis studies by ourselves (Teuber et al., 1977) have also yielded X-ray oscillations of approximately 5-min duration, being associated with flares and containing a few percent of the peak-amplitude power. While the power-spectrum analysis method has proven to be workable and yields significant results, it has the inherent weakness of requiring stationarity to the X-ray data (i.e., the X-ray signal is regarded as an ergodic – continuously periodic – function). Indeed, this may be the case for noise and perhaps even the oscillatory component in flare data; however, the big pulse itself should be treated as a deterministic response (i.e., its profile is the result of some stimulus). Consequently, we report here on a technique which we have applied to the X-REA flare data and have found to be of greater use.

The technique, called principal component analysis, is based on proper or empirical orthogonal expansions (Seal, 1964; Fisher, 1974; Morrison, 1976). While known for quite some time, only a modern computer permits the efficient solution of such eigenvalue problems of a relevant magnitude.

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II. Representation of Solar Flare Data by Empirical Orthogonal Functions

A. The Method

The representation of non-stationary processes by empirical orthogonal functions (i.e., principal component analysis) has been extensively described by Seal (1964) and Morrison (1976), and the technique has been successfully applied to a diversity of research areas; e.g., engineering (Jennings, 1977; Robinson, 1978), atmospheric science (Essenwanger, 1976; Zak, 1977), geophysics (Kjelaas, 1971), and medicine (Cox and Hugenholtz (eds.), 1975; Glaser and Ruchsin, 1976). Our application differs from these previous studies and is believed to be the first application of the technique to full-disk, solar, X-ray proportional-counter data to investigate the flare phenomenon. We have summarized the basis of our principal-component analysis method in the Appendix where we also discuss the step-by-step procedure employed in the program and the limitations of the method.

B. Computational Constraints

The present analysis is based upon programs that evolved from the NATS (i.e., National Activity to Test Software) program and that are compiled under "EISPACK" as Matrix Eigensystem Routines (Smith et al., 1976). The numerical evaluation of eigenvalues from a covariance matrix and subsequent determination of eigenvectors is limited by the length of observational arrays and by the memory size and speed of the available computer to a maximum input-array size of 384 × 384 (elements).

Each analyzed event was recorded by the X-REA in 6 energy channels between 2.5 and 7.25 Å and 4 energy channels between

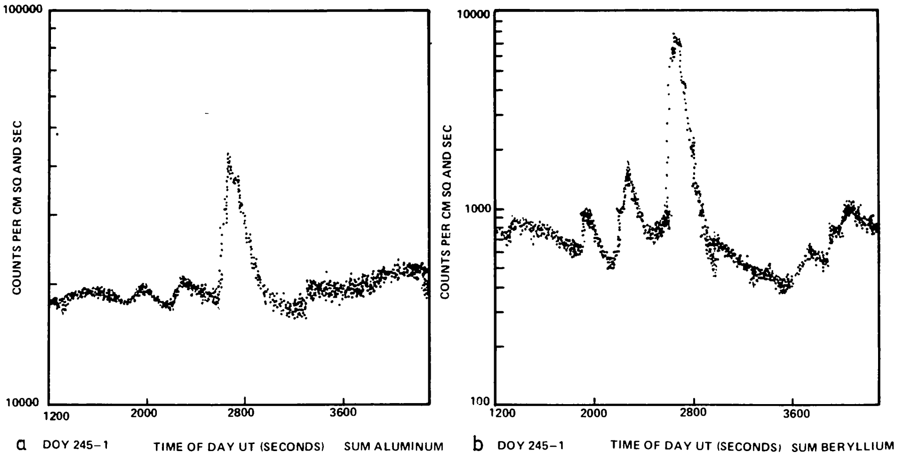

Fig. 1a and b. Raw-data flux profiles versus time for the 2 September 1973 SB/C 4 subflare, showing the sum of aluminum-window counter channel outputs (Fig. 1a) and sum of beryllium-window counter channel outputs (Fig. 1b). One should note the similarity of the two curves

Table 1. Distribution of cumulative power in the first ($E1$) and first and second ($E1+E2$) eigenvectors for several select events

DATE/DOY[a]	OPTICAL/X-RAY FLARE CLASS	TIME OF PEAK EMISSION (UT)	X-REA OBSERVATIONAL WINDOW (UT)	E1(%)	E1 + E2(%)	Δ(%)
15 June 1973 (166-3)	1B/M3	1413	1346-1444	94.7	96.8	2.1
9 August 1973 (221-3)	SN/M1	1553	1506-1600	92.3	95.3	3.0
9 August 1973 (221-4)	SF/C2	2148	2118-2213	94.5	97.8	3.3
2 September 1973 (245-1)	SB/C4	0045	0017-0112	89.9	94.4	4.5
2 September 1973 (245-3)	SF/C2	1620	1550-1644	83.4	91.7	8.3
27 November 1973 (331-1)	SB/MI	0313	0253-0348	95.4	97.5	2.1
2 December 1973 (336-3)	1N/M1	1518	1500-1555	93.6	96.0	2.4
2 December 1973 (336-4)	SN/C5	2015[b]	1940-2035	87.5	94.1	6.6
16 December 1973 (350-4)	SF/C2	1956	1920-2015	93.1	97.2	4.1
15 January 1974 (380-3)	SN/C6	1423	1415-1514	91.5	96.2	4.7

NOTES: a, The numbers in the parentheses under DATE/DOY refer to the Day-of-Year and the particular 6-h block of time within the DOY in which the event occurred; e.g., 166-3 is 3rd quarter of DOY 166, being 1200-1800 UT. Please note that the DOYs continue incrementally past the year change from 1973 to 1974. Thus, 15 January 1974 is DOY 380 not 015.

b, The actual time of peak-level emission, according to Hirman et al. (1977), was not precisely determined. It is logged as having occurred between 2010 and 2020 UT.

6.1 and 20 Å. The data are comprised of samples each 2.5 s in duration and extend up to 55 min in total length. Selected arrays for principal component analysis start at a common, specified initial time with an option for using multiples of 2.5 s as the spacing for data points. As an alternate way of synchronizing the data, each channel can be aligned to a normalized maximum pulse-height. A total of 384 data points represents each channel for an event, where this number is a combination of two numbers to base 2 (i.e., $2^7+2^8=384$) and is a convenient size for subsequent Fast Fourier Transforms of eigenvectors. We note that when 384 arrays are stored in-core with the IBM 360-75, the total program requires a storage capacity of 1.5 million bytes. Also, double precision is required for convergence in the evaluations of the covariance matrix even for a few select roots.

Figure 1, depicting the flux profile of the 2 September 1973 SB/C 4 subflare from Boulder region AR 215 (McMath region 12512) which peaked about 0045 UT, shows a one-dimensional array representing the sums of raw data observed using the X-REA 6.1-20-Å, aluminum-window proportional counter (Fig. 1a) and the 2.5–7.25-Å, beryllium-window proportional counter (Fig. 1b), which is typical of the events analyzed here. A composite input-array for six different events (the first six events listed in Table 1) is shown in Fig. 2, where the events are aligned according to pulse-height. The figure reveals common features of the events in a 384×384 array size, selected so that 38 events can be looked at at any one time. For individual events, our program finds eigenvectors through the ten energy channels of one event and then steps through the same ten energy channels for the next event, and so on.

C. Results

Common features of different events are, without a subsequent normalization (feasible according to Sect. III), more difficult to interpret than common features obtained through all ten channels for each event. These common features are represented in Table 1 by the cumulative power contained in the first (E 1) and first and second ($E1+E2$) eigenvectors and their difference (Δ), in

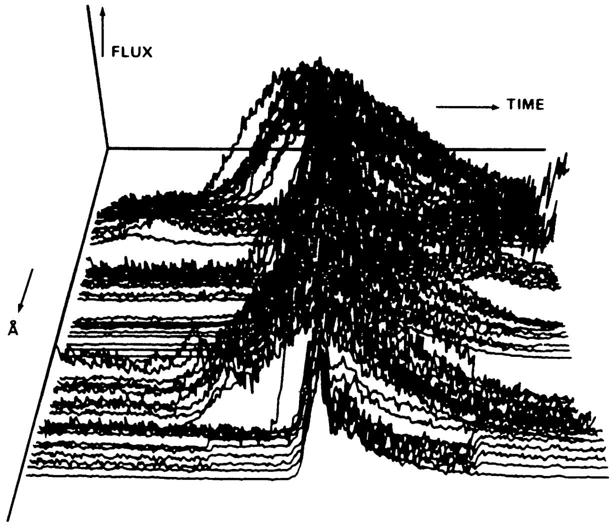

Fig. 2. A composite input-array for the first six events listed in Table 1, illustrating the alignment of X-ray flares according to their pulse-height. A maximum array of 384 × 384 points is possible with our IBM 360-75 computer

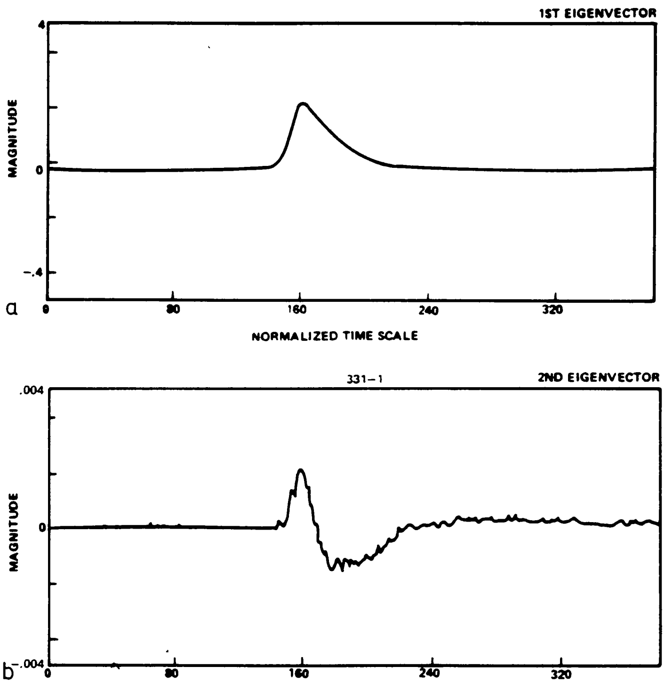

Fig. 3a and b. Displays of the first ($E1$) and second ($E2$) eigenvectors, Figs. 3a and 3b, respectively, for the 27 November 1973 SB/M 1 subflare. The actual magnitude of $E2$ relative to $E1$ is about 2%. One should note the similarity of Figs. 3a and 3b to the analytic curves of Figs. 4a and 4b and the arbitrariness of the phase of the eigenvector in the above displays

terms of percentage, for several select events. The events are listed chronologically by date and DOY (i.e., Day of Year), and the optical and X-ray flare class of each event is identified, as are the overall X-REA observational time and the time of approximate peak emission.

Eigenvectors of the independent variable matrix can be interpreted as that part of the total variance accounted for by the given linear combination of variables where the eigenvector elements are the weights or coefficients. Since the first two components account for roughly 95% of the total variance, the two eigenvectors ($E1$ and $E2$) are considered sufficient to describe the original variables.

We show in Fig. 3 the $E1$ (Fig. 3a) and $E2$ (Fig. 3b) eigenvectors for one of our analyzed events – the 27 November 1973 SB/M 1 subflare from Boulder region AR 287 (McMath 12628) which peaked about 0313—T. The second eigenvector looks, in this event (and in all analyzed events), very similar to the derivative of the first eigenvector. Additionally, all analyzed flares which have been described by orthogonal functions and listed in Table 1 show relationships between the $E1$ and $E2$ eigenvectors which are very similar to those shown in Fig. 3.

III. An Analytic Expression for X-ray Flare Data

A. An Empirical Curve-fit

Since about 90% of the cumulative power of the first eigenvector represents, well enough for practical purposes, each analyzed flare event, an analytic expression for it would be desirable. Many combinations for the constants k, m, b, and n were tried in a function $y = kt^m e^{-bt^n}$ (where y is the X-ray flux and t is time; below in Eq. 1 $t \equiv T$, following the FORTRAN notation). We have found through trial-and-error analysis a χ^2 distribution, satisfactory for an empirical curve-fit, for the $E1$ eigenvector of the observed X-ray flux through all X-REA energy channels. Namely

$$F(x) = F(\chi^2) = K(g)\, T^{(g/2)-1} e^{-T/2}, \tag{1}$$

where $F(x)$ is the X-ray flux, $F(\chi^2)$ is the Chi-square distribution function, g is the number of degrees of freedom which was selected to be 5, T is time, and $K(g)$ is a constant whose value depends strictly on g. The origin of the analytic expression through either an $E1$ eigenvector or the raw data is unknown. Furthermore, we found that it was necessary to match the observed X-ray flux, denoted below as Y, of different flares to an empirical equation, given as

$$Y = A(BT)^{3/2} e^{-BT}, \tag{2}$$

which can be related to a χ^2 distribution with $g = 5$. The hypothesis, then, is that the $E1$ eigenvector of the soft X-ray flux for each event is similar, when matched by an empirical curve-fit.

Using Eq. 2, we have generated templates of the first eigenvector and its derivative for the general, single-impulse-initiated X-ray flare. Figure 4a shows the $E1$ eigenvector normalized in magnitude to 1.0 at normalized time scale $T = 1.5$ and characterizes our standard Chi-square distribution with a fixed origin. Figure 4b depicts the $E2$ eigenvector and is drawn to the same normalized time scale. We note that the extremes in Fig. 4b occur at $T = 0.28$ and $T = 2.72$ and that the absolute magnitude of $E2$ relative to $E1$, for practical curve-fits to the soft X-ray flux, is of only a few percent.

B. General Least-squares Fit

Let $F(a_j + T_i)$ approximate Y_i, representing the X-ray values as a function of time increment $i = 0, 1, 2, 3, \ldots, 384$, where a_j is one of the parameters required for the fitting function of $F(x)$, listed

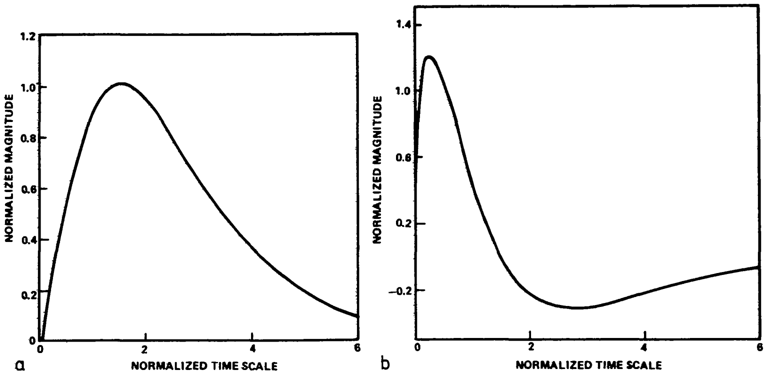

Fig. 4a and b. Displays of the $E1$ eigenvector (Fig. 4a) and $E2$ eigenvector (Fig. 4b). The $E1$ eigenvector has been calculated using Eq. 2; the $E2$ eigenvector is the derivate of the $E1$ eigenvector. One should note that the $E1$ eigenvector has been normalized to unit value $(=1)$ at $T=1.5$ (where T is time), and the $E2$ eigenvector has been normalized to 1.2

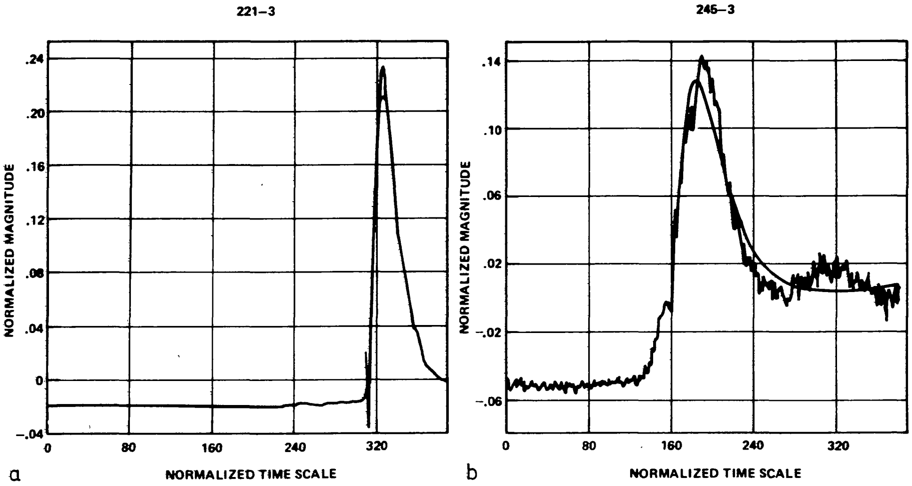

Fig. 5a and b. Superposition of 4-parameter, computed (smooth line) analytic expression, Eq. 4, on computed $E1$ eigenvectors (wavy line) for the 9 August 1973 SN/M1 subflare (Fig. 5a) and the 2 September 1973 SF/C2 subflare (Fig. 5b). The origins of Figs. 5a and 5b were taken as $T=315$ and $T=160$, respectively. For Fig. 5a, the absolute value for the normalized magnitude had to be programmed to the left of the origin in order to avoid computational instability when $T<0$ (in Eq. 2). This manifests itself in the figure (and also in Figs. 7, 9a, and 10) as a slight up-down-up wiggle. The actual sampling rate for Fig. 5a is 1 sample per 7.5 s; for Fig. 5b, it is 1 sample per 2.5 s. The oscillations apparent in the data following the C2 X-ray flare (Fig. 5b) may be a precursor to a subsequent C4 X-ray flare which occurred within the same region slightly later (maximum emission at 1629 UT)

below as $T0(=a_1)$, $Y0(=a_2)$, $A(=a_3)$, and $B(=a_4)$, and T_i is the time. For our least-squares fit we must find parameters, such that

$$\frac{\partial}{\partial a_j}\chi^2=\frac{\partial}{\partial a_j}\sum_{i=0}^{384}\{(Y_i-Y(T_i))^2\}=0, \tag{3}$$

$j=1, 2, 3, 4 \quad j=1, 2, 3, 4$

where Y_i is the X-ray flux (raw data), $Y(T_i)$ is the model function (Eq. 2), and a_j and i are defined as before. The term $(Y_i-Y(T_i))^2$, then, corresponds to the residual between the model function and the real data which must be minimized in order to have a best-fit. We note that at least four parameters are necessary to fit an analytic expression through the computed $E1$ eigenvector; namely, $T0$, $Y0$, A, and B, where $T0$ and $Y0$ are origin values of time and flux, respectively, and A and B are the height and width of the pulse, respectively. The program which we use, similar to the one described by Bevington (1969), starts with four initial guesses which are then iterated until we find a local minimum for Eq. 3 that is determined by the above partial derivatives. The computations converge after a few iterations and yield numerical values for $T0$, $Y0$, A, and B.

For some curve-fits it was necessary to release, after a best-fit to the four parameters, two additional parameters $A1$ and $A2$. These newly introduced parameters account for a superimposed linear term $A1T$ and a quadratic term $A2T^2$ in the analytic expression for the empirical fits given by

$$Y=A[B(T-T0)]^{3/2}e^{-B(T-T0)}+Y0+R, \tag{4}$$

where $A=2.44\,(YM-Y0)$, $B=3/[2(TM-T0)]$, and R is defined as $R=F(T)=A1T+A2T^2+\dots$ The terms YM and $Y0$ refer to maximum and origin values of X-ray flux, respectively. The residuals include full disk, background, X-ray flux variation which is normally quite small and a term representing noise due to the instrument and to telemetry. All fits, then, were satisfied with the aforementioned four parameters; i.e., $T0$, $Y0$, A, and B for the analyzed events, capturing 90% of the data.

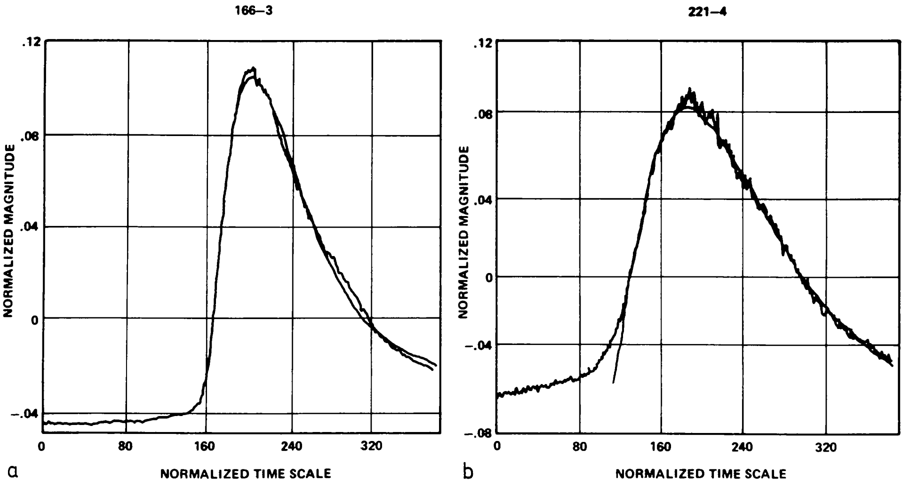

Fig. 6a and b. Superposition of 4-parameter curve-fit (smooth line) on real curve (wavy line) for the 15 June 1973 1 B/M3 flare (Fig. 6a) and the 9 August 1973 SF/C2 subflare (Fig. 6b). One should note that the origin ($T = 159$) falls on the real curve for the M3 event, yet deviates (origin at $T = 112$) for the C2 event, suggesting that additional activity occurs in some X-ray flares which may or may not be related to the big pulse signature

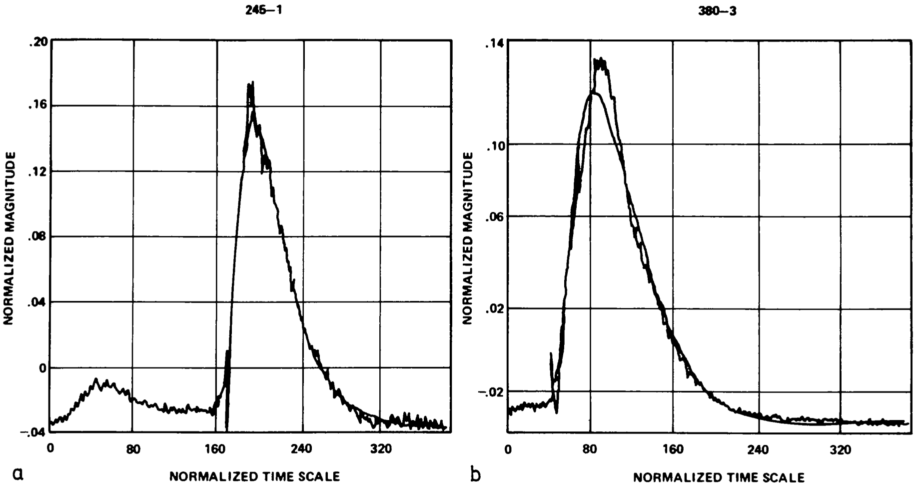

Fig. 7a and b. Superposition of 6-parameter fit (smooth line) on real curves (wavy line) for the 2 September 1973 SB/C4 subflare (Fig. 7a) and the 15 January 1974 SN/C 6 subflare (Fig. 7b). Both event analyses are based on an actual sampling rate of 1 sample per 7.5 s. In Fig. 7a, all 10 X-REA channels were synchronized on maximum pulse height (for the principal component analysis). In Fig. 7b, all 10 X-REA channels were started at an arbitrary and equal time

Figure 5 shows a superposition of a 4-parameter, computed analytic expression according to Eq. 4 on computed $E1$ eigenvectors for different events. In particular, Fig. 5a compares the two curves associated with the 9 August 1973 SN/M1 subflare from Boulder region AR 185 (McMath 12474) which peaked about 1553 UT, and Fig. 5b compares the two curves associated with the 2 September 1973 SF/C2 subflare from Boulder region AR 215 (McMath 12512) which peaked about 1620 UT. Fairly good agreement is observed. In Fig. 6, we compare 4-parameter fits of the two curves to show that occasionally the computed curve origin differs from the real case, suggesting that additional (perhaps, pre-flare) activity occurs which may or may not be related to the big pulse (Fig. 6b – the 9 August 1973 SF/C2 subflare from Boulder region AR 185 or McMath 12474 which peaked about 2148 UT). Likewise, there are occasions when the computed origin is exactly on the big-pulse curve (Fig. 6a – 15 June 1973 1 B/M 3 flare from Boulder region AR 131 or McMath 12379 which peaked about 1413 UT). In Fig. 7, we present a 6-parameter fit of the two curves for the aforementioned 2 September 1973 SB/C4 subflare (Fig. 7a; see also Fig. 1), and a 6-parameter fit of the two curves for the 15 January 1974 SN/C6 subflare from Boulder region AR 314 (McMath 12686) which peaked about 1423 —T (Fig. 7b). Again, fairly good agreement is observed.

C. Curve-fits Through Raw Data

For practical purposes, we have also chosen to use raw data for the empirical curve-fits. A typical example is shown in Fig. 8 (see also Fig 6a – the 15 June 1973 1 B/M 3 flare), where the residuals correspond to the sum of higher-order eigenvectors. By using empirical Eq. 4, we have the means for detrending X-ray flare data and separating them into four significant components: (i) a big deterministic transient that is essentially proportional to $T^{3/2} e^{-T}$ and captures 90% of the signal power; (ii) an oscillatory component which may start prior to the big pulse and modulate it, as well (it is faint and captures only a few percent of the signal power); (iii) a background flux change for the full-disk observation, which may be totally unrelated to the big pulse and the oscillatory components; and (iv) scatter in the data, representing noise as a purely stochastic variable. After numerous curve fits, one observes that the fit for the big pulse for any flare-related event is independent of the energy channel used in the analysis.

IV. Spectral Analysis of Eigenfunctions

In our previous paper (Teuber et al., 1978), we observed an approximate 300-s oscillation using non-detrended data (i.e., the

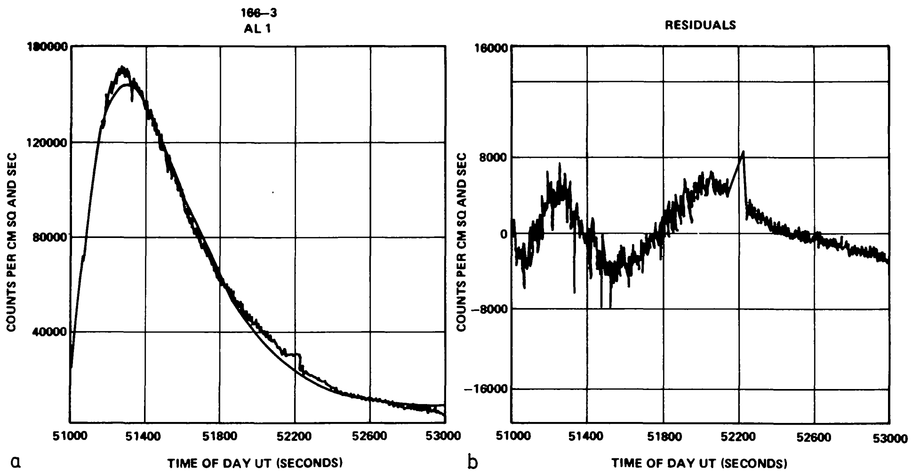

Fig. 8a and b. Superposition of empirical curve-fit (smooth line) through a limited time interval of raw data (wavy line) for the 15 June 1973 1 B/M 3 flare (Fig. 8a). One should note a small data gap occurring near $T=52{,}200$. In Fig. 8a, the term AL 1 means X-REA aluminum-window counter, channel 1 which responds to 16–20-Å X-rays. In Fig. 8b, we show the residuals between the raw data and the analytic curve. Again, the data gap is quite apparent near $T=52{,}200$

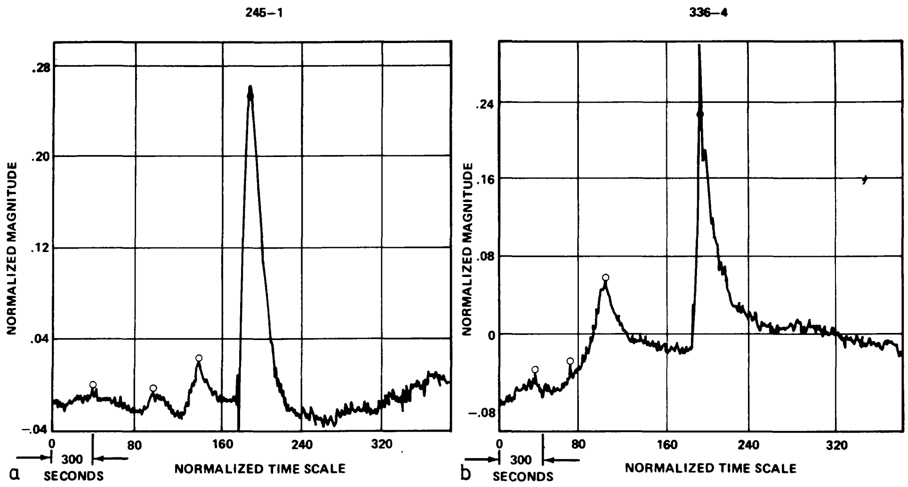

Fig. 9a and b. Display of apparent time-varying X-ray oscillations (period ~5 min), maximum occurrences denoted by small circles (○), in the X-ray data of the 2 September 1973 SB/C4 subflare (Fig. 9a) and the 2 December 1973 SN/C5 subflare (Fig. 9b). Figure 9a has been plotted at the sampling rate of 1 sample per 7.5 s and has an origin at $T=182$. The oscillations are apparent in the curve prior to the origin through $T=240$. Oscillations are also noted in Fig. 9b for the segment $T=0$ to $T=180$. A 300-s reference time-scale is shown for convenience

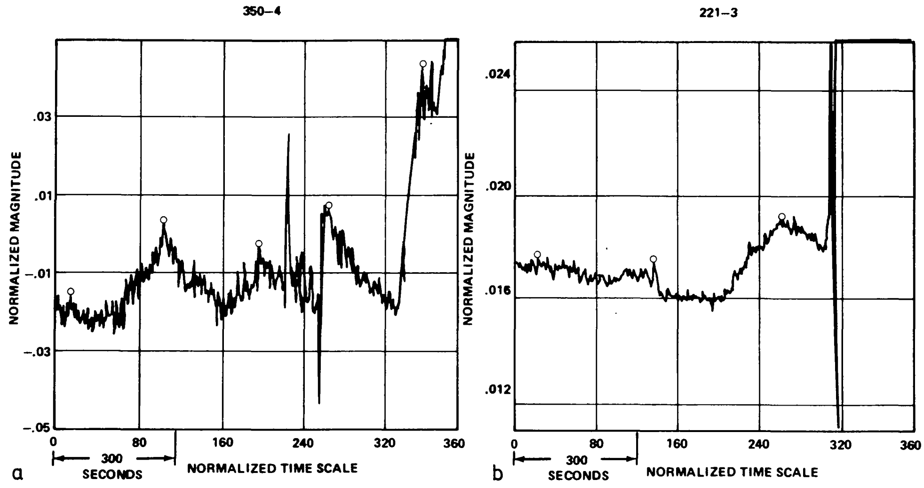

Fig. 10a and b. Displays of apparent time-varying X-ray oscillations (period 5 min), maximum occurrences denoted by small circles (○), in the X-ray data of the 16 December 1973 SF/C2 subflare (Fig. 10a) and the 9 August 1973 SN/M1 subflare (Fig. 10b). We show in Fig. 10a oscillatory enhancements prior to the start of the event at $T=350$ and in Fig. 10b similar enhancements prior to the start of the event at $T=315$. Again, a 300-s reference time-scale is shown for convenience. One should note the two large extraneous points (called outliers in computer terminology), occurring just prior to and after $T=240$

big pulse or transient was not removed prior to analysis) in conjunction with the power-spectrum analysis technique. So, in order to provide a direct comparison to the results of that paper, we have applied a Fourier analysis to the principal components, as well as to non-detrended X-ray flare data. (We note that, while the oscillations are weakly observed in the data, they are greatly enhanced by removing the transient prior to analysis.) Thus, spectral analysis of each of the two largest eigenvectors for different events (see Figs. 9 and 10) suggests the existence of an ~5 min oscillation. However, we note that the eigenvectors, resulting from the generalized least-squares analysis, clearly indicate that the 5-min value is nowhere an exact measure except in a power spectrum

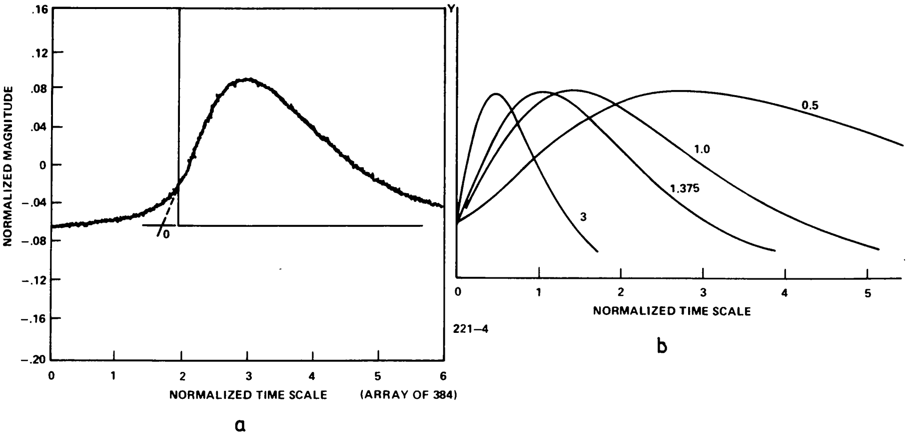

Fig. 11a and b. Superposition of the analog simulation of Eq. 5 (dashed curve) onto the raw data (wavy curve) for the 9 August 1973 SF/C2 subflare (Fig. 11a) and display of a variety of solutions to Eq. 5 for different time constants (Fig. 11b). In Fig. 11a, the time constant is 1.0

which averages over a time period. Figure 9 demonstrates this by showing two events – the aforementioned 2 September 1973 SB/C4 subflare (Fig. 9a) and the 2 December 1973 SN/C5 subflare from Boulder region AR 292 (McMath 12628) which peaked between 2010 and 2020 UT (Fig. 9b) – that display irregularly shaped and displaced oscillations before and possibly after the respective primary events (big pulses). A 300-s time-scale is shown for each event as a reference. Likewise, Fig. 10 shows similar findings for the two events occurring on 16 December 1973, an SF/C2 subflare from Boulder region AR 300 (McMath 12664) which peaked about 1956 UT (Fig. 10a), and on 9 August 1973, the afore-mentioned SN/M1 subflare. However, unlike those events of Fig. 9, the events of Fig. 10 only show oscillations before the event. (The events occurred near the end of an X-REA observing period and, hence, were not well observed over their total duration.) Again, a 300-s time-scale is drawn for convenience.

While in earlier years it was held that the observed (photospheric and chromospheric wavelengths) oscillations may indeed relate to cyclic waves in the chromosphere-photosphere (e.g., Zirin, 1966), i.e., likened to oceanic waves with a natural sloshing period, recent studies have essentially demonstrated that the 5-min oscillations are high-order, non-radial (p-mode) acoustic waves of the Sun (see reviews by Deubner, 1977, and Leibacher, 1977). Thus, the X-ray oscillations reported here may somehow be related to the excitement of these high-order, non-radial acoustic waves in the corona.

V. The Soft X-ray Flux of a Flare as a Deterministic Response to an Impulse

In the preceding sections, we have treated the big-pulse X-ray flare as a purely deterministic response to an impulse. If we consider Eq. 2 as the solution of a differential equation with a time-varying parameter, namely,

$$\ddot{Y}+2\dot{Y}+\left(1-\frac{3}{4t^2}\right)Y=0, \tag{5}$$

where Y is defined as in Eq. 2, and t equals BT, then we see that the solution is unstable for $t \leqq (3/4)^{1/2}$ and stable for $t>(3/4)^{1/2}$. For $t=0$, the coefficient for Y is $-\infty$ and represents a singularity a condition which may be supportive of a result reported by Falciani et al. (1977) concerning the existence of a single triggering-mechanism during the flash phase of a solar flare. (We note that the general solution for Eq. 5 is according to Durrant, private communication, $Y=\lambda t^{-1/2}e^{-t}+\mu t^{3/2}e^{-t}$, where Y and t are defined as before and μ and λ are simple scale factors. Analog simulation shows that, in our analysis of X-ray flux, $\mu \gg \lambda$. Thus, Eq. 2 is a solution to Eq. 5 when $\lambda=0$.)

Figure 11a depicts a superposition of analog and digital (raw data) simulations, where the analog simulation is a solution to Eq. 5 for the afore-mentioned 9 August 1973 SF/C2 subflare. The curves lie one atop the other. Figure 11b displays a variety of similar solutions to Eq. 5 for different time constants. Thus, a time constant could be selected for an event such that the analog solution would describe the raw data exactly (as in Fig. 11a, where $t=1.0$). This implies that the big-pulse X-ray flare is purely a deterministic response to an impulse (Dirac), and, likewise, suggests that it can be predicted from a small data set (e.g., the first 10–20% of the total data set during flare rise). In fact, we have used such a scheme to reveal flare peak-level emission, time of peak-level emission, and event duration for selected X-ray flares (Teuber et al., 1979). There is from the practical point of view no need to solve the differential Eq. 5 for empirical curve-fits. The analytical expression (Eq. 4) is much easier to use; however, it does not point to the concept of an impulse-response function for further analysis.

For some flares the measured flux deviates appreciably from the computed flux near the origin (T 0, Y 0). We suggest that this may be caused by nonflare-related flux stemming from the full-disk X-ray observations (i.e., the brightening of other active regions or the emergence of new flux). However, we also note that for some flares, and in particular for those of big flux-peak values, the deviation between observed and computed values at the analytic origin is extremely small. Consequently, the uncertainty for the time of the triggering impulse for Eq. 5 is within the X-REA sampling rate of one sample per 2.5 s (i.e., the impulse duration may be $\leqq 2.5$ s). This condition was verified by an analog simulation of Eq. 5.

Exponential decays of soft X-ray flares have been the topic of many previous studies (e.g., see Švestka, 1976, and references contained therein). However, little has been expounded regarding the treatment of X-ray flares as purely deterministic responses. We emphasize that the analytic curve-fit equations (Eqs. 2 and 5) trace the X-ray flux continuously with little deviation from a suspected origin as an impulse-response function throughout the event duration or for as long as the X-REA observational window permitted. In more than twenty analyzed events the analytic solution (Eq. 2) of the differential Eq. 5 was found to converge to the observed (raw) data only a few seconds away from the computed origin.

VI. Conclusions

Principal component analysis has been used in a study of solar X-ray flares. The technique emphasizes the existence of time-varying oscillations in the solar soft X-ray flux and is useful for separating noise from the signal of proportional-counter data and for extracting deterministic components of X-ray flares. We find that the first two eigenvectors in all of the analyzed events contain approximately 95% of the cumulative power in the signal, a condition which is sufficient to describe the events in detail.

An analytic expression (Eq. 2) and a corresponding differential equation (Eq. 5) with a time-varying parameter resulted from our study of the principal-component analysis method. The differential equation points to an impulse as the physical triggering mechanism of the big pulse in the soft X-ray spectrum of a solar flare. (Mathematically, the expressions point to an initial singularity.) We note that a precise determination of the origin from a generalized least-squares fit to the proportional-counter data is frequently obscured due to smaller, background, X-ray flux variations and due to the uncertainty of the impulse within the 2.5-s sampling duration of the instrument.

The present study suggests that the big X-ray pulse is essentially a deterministic and predictable response; i.e., the result of an impulse. Apparently, unrelated events appear to be similar to each other when normalized by time of peak-level emission using two parameters: A (pulse height) and B (pulse width). We find that a generalized least-squares fit can be applied to either the first eigenvector of the ten channels of the X-REA or directly to the raw data from each channel. In the fits through the raw data, any one channel is sufficient for capturing about 90% of the cumulative power in the signal. Furthermore, if one neglects variations within about 10% of the signal, then the channels between 2–20 Å are virtually identical and can be considered to stem from a monochromatic source. That all analyzed flares look quite similar in the $E1$ and $E2$ eigenvectors is not really too surprising from the fact that the X-REA looked at the full-disk Sun while flares themselves are known to originate from much smaller point-like sources (in comparison). Hence, the X-REA observed only temporal changes, leaving the spatial changes unresolved.

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Appendix A

This Appendix serves as a supplement to our previous textual discussion (Sect. II.A), outlining the aims, assumptions, and limitations of the method as applied to the X-REA data and ending with a step-by-step procedural overview of our program.

A.1. Aims

Our data consist of a set of waveforms and it is of interest to analyze the nature of the relationships that exist between the set members. It is our aim to discover attributes that they share in common. If, furthermore, the number of basic waveforms needed to represent adequately all the observed waveforms is substantially less than the number of data waves, we can both achieve a significant degree of data reduction and simplify a study of the waveform interrelationships. Our data reduction then serves as a basis for a subsequent modeling effort, as shown by a derived differential equation (Eq. 5).

A.2. Assumptions

If the basic waveforms can be given physical meaning by relating them, for example, to the Boltzmann collision integral or the Fokker-Planck equation, then such a representation can provide direct insight into the plasma-physical mechanisms generating the data. A necessary condition for a linear representation of the data set by principal components to serve as a framework for a model is that, to within a reasonable approximation, the solar generators which give rise to the data interact in a linear manner (e.g., the relaxation time for an observed event being proportional to the ratio of density and temperature). We cannot at this time describe the physical parameters for our analyzed data within the context of the principal component analysis method. However, it is this final goal which would make the disproportionate effort of the data reduction by principal components really worthwhile.

A.3. Limitations

A drawback in utilizing the covariance as the similarity measure is that strong components within the waveforms will have the greatest effect upon the value of the correlation. Consequently, there is a danger that important, but low-magnitude details, may be obscured by a principal component analysis (e.g., a 3-min oscillatory component). We try to avoid this by adding a spectral analysis (FFT) to the eigenvectors $E1$ and $E2$.

In general, there is no mathematically unique linear representation of the X-REA data, no matter how concise the representation may be. The choice of our particular representation is, to some extent, arbitrary. Consequently, such analyses by principal components are not expected to lead to totally new discoveries, but rather they might better quantify and enhance an objective description of previously partially understood phenomena. Our choice of a Chi-square distribution gleaned from the principal component analysis can perhaps result in a meaningful expansion and can be related to solar plasma physics.

A.4. Procedural Steps

Several steps must be accomplished in our principal-component analysis program. First, we must perform a *time normalization* of the data. The input arrays are based either on fixed start times or an alignment of maximum pulse heights (of events). Both methods have been tried and show no significant differences in the resulting eigenvectors nor their cumulative power.

Next, we must perform a *2-dimensional input-array selection*, where the input array represents the matrix A_{mn}, m being the number of rows (degrees of freedom) of the matrix, i.e., the number of X-REA energy channels (nominally, $m=10$ for each event), and n being the number of columns of the matrix, i.e., the number of time increments that the event has been broken-up into (nominally, $n=384$; the actual amount of time per increment equals either 2.5 s, the time resolution element of the X-REA, or some multiple of 2.5 s). We perform this 2-dimensional input-array selection to provide insight for common features when comparing several events simultaneously. For a single event, m equals 10 and we compute 10 eigenvalues. For several events we can compute qm eigenvalues, where qm is the number of events (q) being studied times the number of energy channels (m) per event. The maximum number of eigenvalues (for an integral number of events) that can be computed in our program is 380 (or 38 events, each event being analyzed in all 10 energy channels).

Third, we must perform a *magnitude normalization* (flux). We do this (for each channel) by means of a computational routine that removes the DC component from each data vector (channel) and also scales the input data so that the maximum pulse height above the mean is of unit value (i.e., equal to 1).

Fourth, we must perform a *matrix/vector sums generation*. Again, this is accomplished by means of a computational routine that generates a matrix of sums-of-squares and a vector-of-sums from the afore-mentioned data vector (see the previous step). This routine allows the generation of a correlation/covariance matrix, performed below as our subsequent step.

Fifth, we must perform a *covariance matrix computation*. This is achieved mathematically by means of the formula

$$C=\frac{1}{m-1} A^T_{mn} A_{mn}, \tag{A-1}$$

where C is the covariance matrix, A^T_{mn} is the transposed matrix of A_{mn}, and m and n are defined as before. Originally, we consider in our analysis that the different channels are dissimilar in character. In this case no physical interpretation is placed on the relative values of the diagonal elements of the covariance matrix. However, subsequent analysis, using an empirical curve-fit (Eq. 2) through the X-REA data (i.e., each proportional-counter energy channel), revealed that all 10 X-REA channels are, indeed, similar when 90% of the cumulative power in the signals is sufficient for their representation.

Sixth, we must perform the EISPACK *routine*. The EISPACK routine gives a complete eigensolution to the covariance matrix. The eigenvectors are first normalized and then arranged (as are the eigenvalues) in descending order. This is readily observed in Fig. 2.

Seventh, we must perform an FFT on the previous step. The FFT allows us to ascertain time-varying periodicities in the soft X-ray flux. Indeed, we see an approximate 5-min oscillation in the first two eigenvectors ($E1$ and $E2$) of our analyzed events (e.g., see Figs. 9 and 10).

Last, we must perform a *cumulative, percent-signal-power computational routine*. This routine computes the cumulative percent-signal-power in a set of empirical orthogonal-function vectors.

Statistical tests of the principal components are only possible for large samples ($q \gg 10$, where q is the number of analyzed events). Hence, because of our small sample size ($q=10$), we did not attempt any statistical tests. We use the principal-component analysis technique (with arbitrary procedures in data normalization and energy channels), therefore, only in the context of an exploratory tool. As such, we found that the technique provided us the clue to a Chi-square fit through the raw data.

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